

Health Effects to Dietary Sodium Reduction

Technical review of the source-split CVD model and preliminary results for prioritized countries

Resolve to Save Lives (RTSL) and University of Washington

August 2026

Recap and today's plan

Recap: where we left off (July 23 call)

From the **July 23** RTSL–UW modeling call, three directions were set:

- **LSS** — UW had presented two options: (1) apply the **SSaSS trial relative risks** directly, or (2) model **sodium + potassium** → **SBP** → **CVD** mechanistically (following Matti and colleagues). Applying SSaSS RRs to the *general* population was judged inconsistent (trial participants were higher-risk), and option (1) cannot represent the whole-population scenario. **Agreed:** investigate and implement option (2) so scenario 2 can be modeled.
- **Fiscal policy** — keep a **simple, transparent** Bloomberg approach; use the **Philippines study** as the primary reference; add **sensitivity analyses** for the large parameter uncertainty. Fiscal is the **lowest-priority** lever for the reinvestment inputs.
- **Euromonitor** — replace the single global \$2.6%/yr packaged-growth assumption with **country-specific** trends (Matti to supply).

Today we present the **current build** — the work completed since July 23 — and **preliminary results** for the 10 prioritized countries.

Today's agenda

1. **What's changed since July 23** — the new mechanistic **Na/K LSS model** (now primary; SSaSS a benchmark), the explicit **fiscal tax**→**price**→**demand pathway** (exploratory, with low/base/high sensitivity), and **country-specific Euromonitor** packaged-food trends.
2. **Model inputs** — updated Euromonitor packaged-food trends, the five-source sodium split, and the applied salt-effect (η) by intervention.
3. **Methods** — the source-split logic (sources $\rightarrow \eta$), the current **Na/K LSS model** (and how it compares with Huang et al. 2026), and the **fiscal pathway** (vs Saxena et al. 2025).
4. **Preliminary results** — baseline burden, projected trajectories by scenario, and deaths delayed across the 10 prioritized countries.
5. **Limitations & discussion** — remaining open items and next steps.

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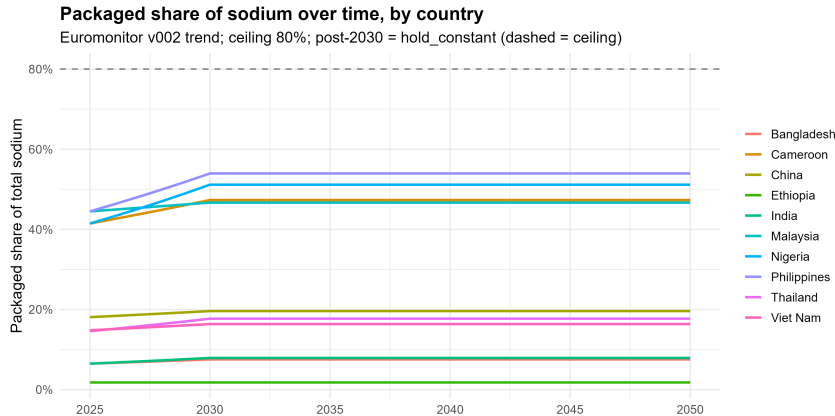
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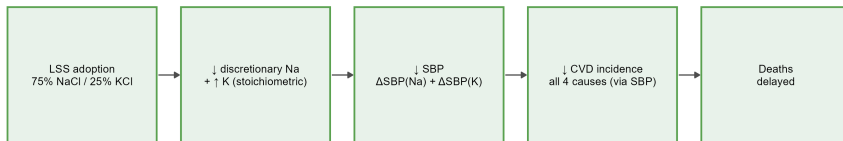
Methods

Packaged share of sodium over time

Each country's packaged share grows on its own Euromonitor trend to a 2030 end-state, then holds flat (dashed line = the 80% ceiling). The other four sources renormalise each year — a composition shift only.



LSS (primary): mechanistic Na/K → SBP → CVD



Central equation: $\Delta\text{SBP}_{\text{LSS}} = \Delta\text{SBP}_{\text{Na}} + \Delta\text{SBP}_{\text{K}}$

- **Na:** 75% NaCl / 25% KCl substitute cuts discretionary Na (KCl × adherence 0.92 × uptake 1.00 = 23%); Filippini Na→SBP 2.8/1.0 mmHg per g (raised/normal BP); 2 g/day floor.
- **K:** added K → excretion (39.1 mg/mmol; ÷1.3) → Filippini U-curve (Filippini, Naska, et al. 2020); baseline-Na modulated; ×0.50 in normotensives; baseline K Reddin 2023 (Reddin et al. 2023).
- **Combine:** Na+K additive ⇒ all four CVD causes; case fatality unchanged.

Coverage scenarios

- **s2** whole population (coverage 50%)
- **s4** diagnosed HTN — NCD-RisC 2019 (15% of adults)
- **s5** treated HTN — NCD-RisC (10%)

Mean ΔSBP (s2) = 1.27 mmHg (= 0.33 Na + 0.94 K).

Benefit-only: harms none (no CKD/hyperkalaemia channel). SSaSS trial RRs are an s4/s5 *benchmark* only — never combined.

How our LSS model compares with Huang et al. (2026)

Same **Na + K** → **SBP** structure, 75% NaCl / 25% KCl formulation, Filippini (2020) potassium dose–response and Reddin (2023) baseline potassium — but narrower in scope and **benefit-only**.

Component	Huang et al. 2026	Current UW/RTSL model
Modelling approach	Static comparative risk assessment (SBP-distribution shift); one time point	Dynamic Well→Sick→Dead state-transition; annual projection to 2050
Geographic scope	193 countries/territories	10 RTSL priority countries
LSS formulation	75% NaCl / 25% KCl (15/85 in sensitivity)	75% NaCl / 25% KCl
Sodium → SBP	Mozaffarian 2014 meta-regression, by age and hypertension status; WHO 2 g/day floor	Filippini Na→SBP (2.8 mmHg/g raised-BP; 1.0 mmHg/g normal-BP); WHO 2 g/day floor
Potassium → SBP	Filippini 2020 non-linear; stronger at higher baseline sodium; additive (80% in a sensitivity)	Filippini 2020 achieved-excretion U-curve; baseline-Na modulation; non-HTN ×0.50; additive
Baseline potassium	Reddin 2023 (global mean ≈ 2.3 g/day)	Reddin 2023 (country × sex; region-mean fallback: Viet Nam, Ethiopia)
Population scenarios	5: all salt / discretionary / non-discretionary / diagnosed HTN / treated HTN	3: whole-population discretionary (s2), diagnosed HTN (s4), treated HTN (s5) — Huang's 2/4/5
Hypertension targeting	NCD-RisC 2019; 80% uptake (20% excluded for hyperkalaemia risk)	NCD-RisC 2019 diagnosed/treated population shares; uptake 1.00 × adherence 0.92
CVD outcomes	Broad CVD basket (IHD; ischaemic, intracerebral & subarachnoid stroke; HHD; AF; PAD ...) + CKD	4 causes: IHD, ischaemic stroke, intracerebral haemorrhage, HHD (all via SBP)
CKD / hyperkalaemia harms	Modelled as a NET effect for population-wide scenarios (benefit – hyperkalaemia harm)	none (no CKD/hyperkalaemia channel) — benefit-only
Uncertainty	Probabilistic (1000 Monte Carlo) + 11 deterministic sensitivity analyses	Point estimates; SSaSS trial-RR as a sensitivity benchmark; input CIs shown, not propagated
Role of SSaSS trial	Composition & adherence only (NOT the SBP effect size)	Optional direct stroke-RR benchmark (HTN scenarios only); separate pathway, never combined

Key departures: LSS is integrated in a *dynamic* 10-country CVD model (not a static 193-country CRA), covers four SBP-mediated causes (not Huang's broad CVD + CKD basket), and does **not** model the CKD / hyperkalaemia harm channel Huang includes for population-wide scenarios. These are differences in scope, not corrections (Huang et al. 2026).

LSS: Huang et al. (2026) vs. our current implementation

Huang et al.: static comparative risk assessment

$$Na^1 = Na^0 - 0.25 Na^R, \quad K^1 = K^0 + \Delta K$$

$$\Delta SBP = \Delta SBP_{Na} + \Delta SBP_K$$

Post-LSS SBP distribution is shifted:

$$X^1 \sim N(\mu + \Delta SBP, \sigma^2)$$

Disease impact is calculated using the full RR function:

$$PIF = 1 - \frac{E_{P1}[RR(X)]}{E_{P0}[RR(X)]}$$

$$D^{averted} = PIF \times D^0$$

Net effect also subtracts hyperkalaemia harm:

$$D^{net} = D_{SBP}^{averted} - D_{CKD}^{hyperK}$$

UW-RTSL: dynamic state-transition implementation

LSS displaces discretionary sodium:

$$\Delta Na = Na^0 q_D(0.25)(0.92)(u)(c_t)$$

and adds potassium stoichiometrically:

$$\Delta K = \Delta Na \frac{0.524}{0.393} \approx 1.33 \Delta Na$$

SBP reduction is:

$$\Delta SBP_{Na} = [2.8H + 1.0(1 - H)] \Delta Na$$

$$\Delta SBP = \Delta SBP_{Na} + m_{Na} m_H \Delta SBP_K$$

where ΔSBP_K comes from the Filippini K-excretion curve.

Rather than a PIF, ΔSBP modifies annual incidence:

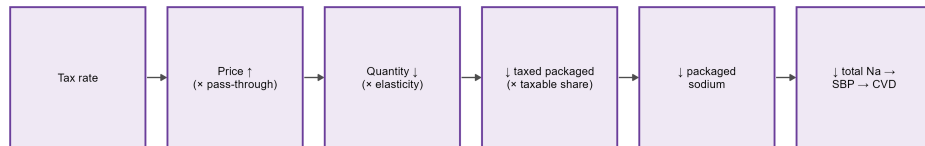
$$IR'_{b,c} = IR_{b,c} \left[1 - (1 - RR_{b,c}^{10}) \frac{\Delta SBP}{10} \right]$$

and the new incidence enters the annual population/state-transition model:

$$Pop_t \rightarrow Cases_t \rightarrow Deaths_t$$

CF unchanged: $CF' = CF$.

Fiscal policy (exploratory): explicit tax pathway



Base case (worked): 20% tax $\times$ 100% pass-through = 20% price rise; $\times$ elasticity -0.50 $\Rightarrow$ **10%** fewer taxed products; $\times$ **50%** taxable-packaged share $\Rightarrow$ **5.0%** of packaged sodium; then $\times$ each country's packaged share of *total* dietary sodium $\rightarrow$ SBP $\rightarrow$ CVD. **Tax rate $\neq$ sodium reduction; not all packaged sodium is taxable.**

Variant	Tax	Pass-thru	Price Δ	Elasticity	Qty response	Taxable share	Substitution	Reformulation	Δ taxed packaged	Δ total packaged
fiscal_low	10%	80%	8%	-0.30	-2.4%	40%	30%	0%	1.7%	0.7%
fiscal_base	20%	100%	20%	-0.50	-10.0%	50%	0%	0%	10.0%	5.0%
fiscal_high	20%	100%	20%	-0.80	-16.0%	60%	0%	5%	21.0%	12.6%

Table M5 (low/base/high). The last column, " Δ total packaged", = " Δ taxed packaged" $\times$ taxable share. Parameters are RTSL-to-confirm.

Fiscal policy: our model vs Saxena et al. (Philippines, 2025)

Saxena et al. (Saxena et al. 2025) is our conceptual and empirical **anchor** — a 20% tax giving ≈100 mg/day (≈10%) sodium over 20 years. Ours deliberately **simplifies** that structure into a transparent, transferable multi-country approximation.

Component	Saxena et al. 2025 (Philippines)	Current UW/RTSL fiscal model
Setting & model	Philippines; proportional multistate life-table (Markov) cohort	Multi-country; embedded in the shared Well→Sick→Dead state-transition engine
Time horizon	20 years (results at 5/10/20 y)	Annual to 2050 (scale-up 2026–2030)
Tax rate	20% excise (40% in sensitivity)	base 20% (low/base/high 10% / 20% / 20%)
What is taxed	WHO WPRO nutrient-profile thresholds; 17 packaged categories; sodium + sugar	Flat taxable-packaged share (base 50%); NO item-level nutrient profiling
Pass-through	100% base (50% / 150% in sensitivity)	base 100% (low/base/high 80% / 100% / 100%)
Demand response	QUAIDS demand system; own- AND cross-price elasticities	Single own-price elasticity (low/base/high -0.30 / -0.50 / -0.80); NO cross-price terms
Wealth / equity	Wealth-quintile-specific elasticities; equity outcomes reported	None — a single national response (no wealth stratification)
Substitution & reformulation	Within-category taxed↔untaxed substitution; endogenous threshold-distance reformulation	Flat offsets only (base substitution 0%, reformulation 0%; varied in low/high)
Packaged-food trend	Euromonitor 2008–19, dynamic; total intake held via offset assumption	Euromonitor packaged trend; composition-only, total intake held
Health pathway	Sodium→SBP→CVD AND sugar→BMI→type-2 diabetes	Sodium→SBP→CVD only (no sugar / BMI / diabetes pathway)
Diseases modelled	Type-2 diabetes, IHD, ischaemic stroke (incidence only)	IHD, ischaemic stroke, intracerebral haemorrhage, HHD (incidence only)
Sodium reduction	≈ 100 mg/day (≈ 10% relative) at year 20	base ≈ 10% of taxed packaged → 5.0% of packaged sodium
Economic evaluation	Full: HALYS, health-care savings, tax revenue (~US\$13 B), ICER, equity	Health impact only — no costing, tax revenue or ICER (yet)
Uncertainty	Probabilistic (1000 Monte Carlo) + 5 deterministic scenarios	Deterministic low / base / high variants only

Not reproduced (by design): the full QUAIDS demand system; category-specific own/cross-price elasticities; wealth-quintile heterogeneity; item-level tax eligibility; endogenous reformulation; the sugar→BMI→diabetes pathway; and tax-revenue / full economic evaluation.

Fiscal policy: Saxena et al. vs. current RTSL model

Saxena et al. (Philippines)

$$Tax \rightarrow Price_j \rightarrow Q_j \rightarrow Na \rightarrow SBP \rightarrow CVD$$

****Tax → food demand****

Uses a multi-product demand system with own- and cross-price elasticities:

$$\frac{\Delta Q_i}{Q_i} \approx \sum_j \epsilon_{ij} \frac{\Delta p_j}{p_j}$$

This explicitly captures substitution between taxed and untaxed foods.

****Food demand → sodium****

$$Na^{tax} = \sum_j Q_j^{tax} Na_j^{tax}$$

Sodium changes arise from both:

- consumer demand response - product reformulation

****Sodium → health****

Changes in sodium shift the SBP distribution; PIFs translate this shift into disease incidence.

Key distinction: Saxena models food demand and substitution explicitly using QUAIDS; RTSL uses a parsimonious, transferable approximation requiring fewer country-specific inputs while preserving the pathway tax → consumption → sodium → SBP → CVD.

Current RTSL implementation

$$Tax \rightarrow Price \rightarrow Packaged Na \rightarrow Total Na \rightarrow SBP \rightarrow CVD$$

****Tax → packaged sodium****

We approximate the packaged-sodium effect as:

$$\eta_{pkg} = \theta_{taxable} [|\epsilon| \tau \pi (1 - s) + r_{reform}]$$

where the parameters represent taxable share, elasticity, tax rate, pass-through, substitution, and reformulation.

****Packaged → total sodium****

$$\Delta Na_{c,t} = Na_c^0 \left(Share_{c,t}^{pkg} \eta_{pkg} \right)$$

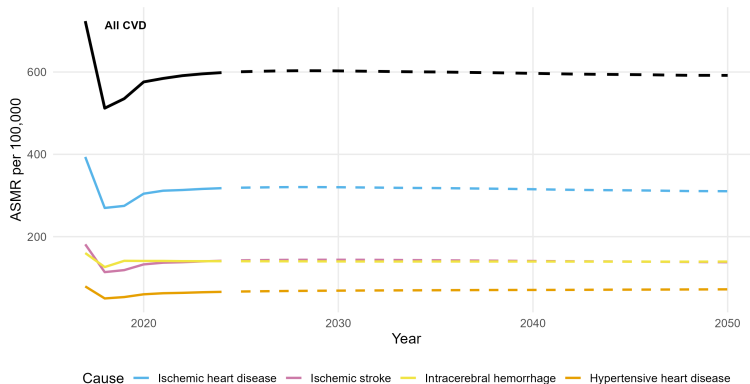
Country-specific packaged-food shares determine the total sodium effect.

****Sodium → health****

The resulting sodium reduction shifts SBP and is propagated through the existing cause-specific CVD model.

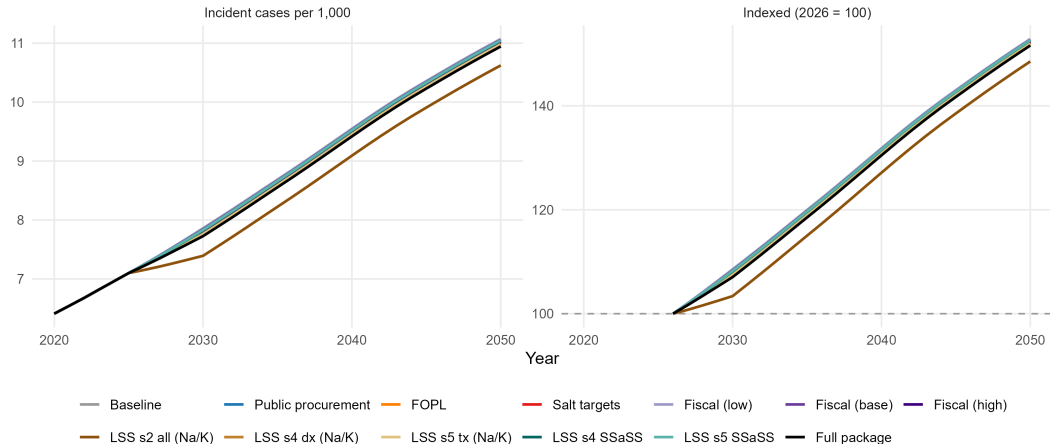
Baseline and projections

Baseline CVD burden, 2017–2050



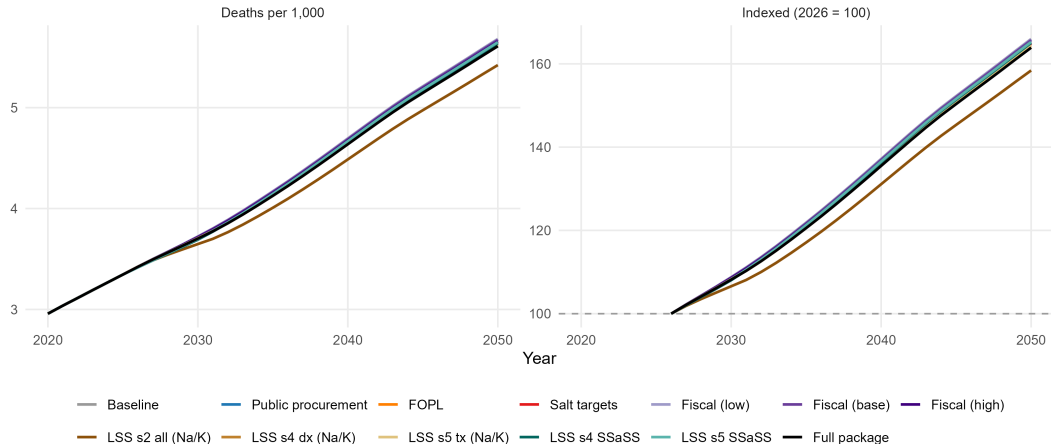
Age-standardised mortality (WHO standard, ages 20–95+) declines with forecasted background-mortality and case-fatality trends; absolute CVD deaths still rise with population growth and ageing. Solid = historical, dashed = projected.

Projected incidence by scenario



Incident CVD cases per 1,000 (left) and indexed to 2026 = 100 (right). Scenarios coincide until scale-up begins, then fan out; the **full package** falls fastest. **Fiscal now separates from baseline** (applied $\eta = 5\%$ on packaged, provisional).

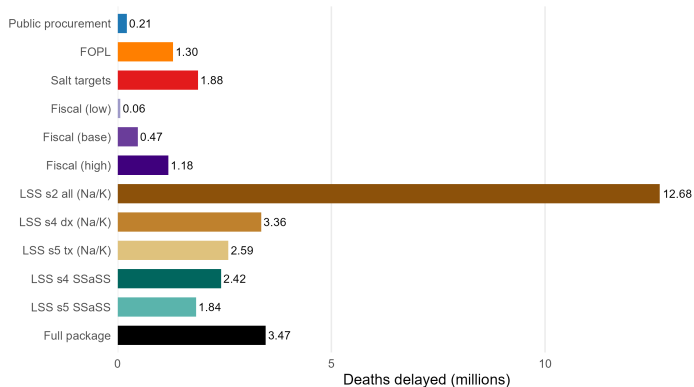
Projected deaths by scenario



CVD deaths per 1,000 (left) and indexed to 2026 = 100 (right). Because mortality responds to the **prior** year's incidence, the scenarios' divergence lags the incidence divergence by about a year.

Main results

Deaths delayed by scenario, 2026–2050



Headline

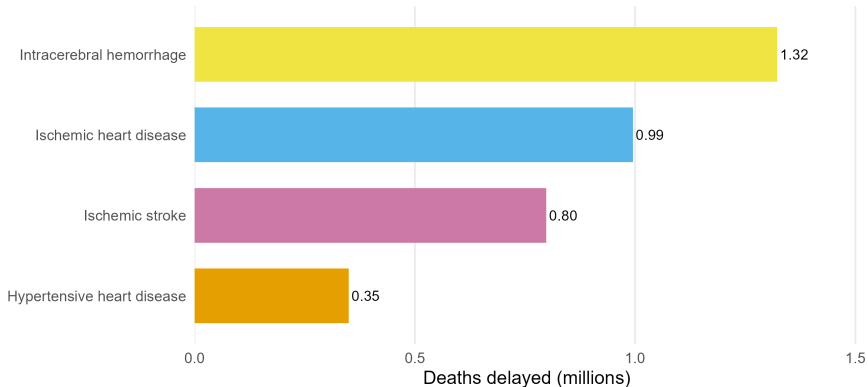
The **full package** delays **3.47 million** CVD deaths across 10 countries (1.0% of baseline CVD deaths in the window; ~139 thousand/year). **full package does not include LSS**

Deaths delayed by scenario — detail

Scenario	Deaths delayed (millions)	Avg per year (thousands)	% of baseline
Public food procurement & service	0.214	8.6	0.06%
Front-of-pack labelling (FOPL)	1.297	51.9	0.37%
Salt targets (reformulation)	1.881	75.3	0.54%
Fiscal policy - low (EXPLORATORY salt tax)	0.063	2.5	0.02%
Fiscal policy - base (EXPLORATORY salt tax)	0.470	18.8	0.14%
Fiscal policy - high (EXPLORATORY salt tax)	1.184	47.4	0.34%
LSS - whole-population discretionary (Na/K-SBP)	12.683	507.3	3.65%
LSS - diagnosed hypertension (Na/K-SBP)	3.359	134.4	0.97%
LSS - treated hypertension (Na/K-SBP)	2.591	103.6	0.75%
LSS - diagnosed hypertension (SSaSS trial-RR benchmark)	2.418	96.7	0.70%
LSS - treated hypertension (SSaSS trial-RR benchmark)	1.836	73.4	0.53%
Full package (public_procurement + fopl + salt_targets + fiscal)	3.465	138.6	1.00%

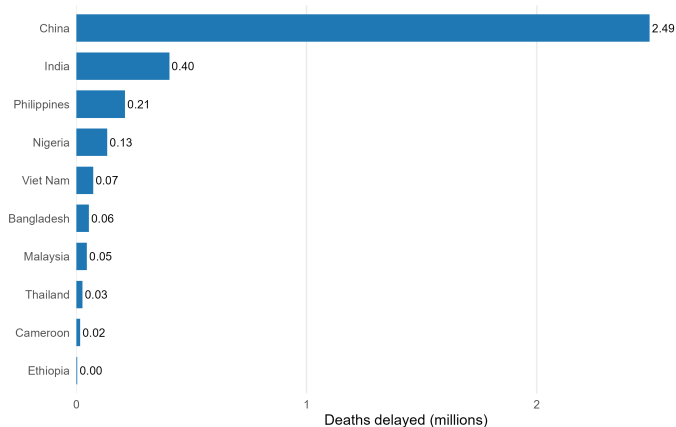
The **LSS (Na/K)** scenarios are the largest single levers — whole-population **s2** 12.68 M, diagnosed-HTN **s4** 3.36 M, treated-HTN **s5** 2.59 M — because they cut discretionary salt (dominant in the most populous countries) through **both** sodium and potassium. Among the source-split policies, **salt targets** delays the most (1.88 M), then **FOPL** (1.30 M), **public procurement** (0.21 M, provisional) and **fiscal (base)** (0.47 M, exploratory). The **SSaSS trial-RR benchmark** (stroke-only) gives **s4** 2.42 M / **s5** 1.84 M — lower, as expected for a stroke-only pathway. Media is excluded.

Full package: deaths delayed by cause



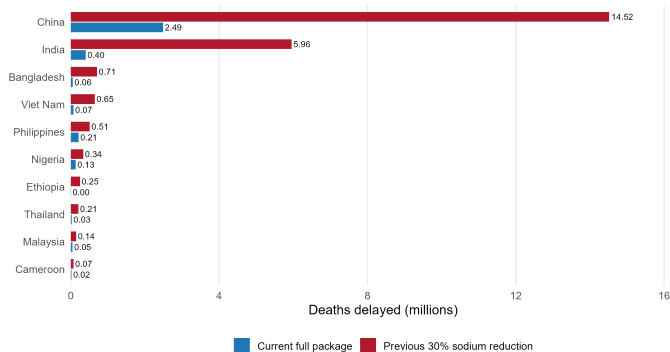
Ischemic heart disease and stroke dominate the delayed deaths, tracking their share of the baseline CVD burden and their SBP-mediated relative risks.

Full package: deaths delayed by country



Absolute counts are dominated by the most populous countries (China, India); per-capita impact depends on baseline burden and local source structure.

Benchmark: full package vs. the 100M-Lives 30% sodium cut



Contextual cross-check

Across the 10 selected countries, the current **full package** delays **3.47 M** CVD deaths vs. **23.36 M** for the previous **100 Million Lives Saved** flat 30% population sodium reduction ($\approx 0.15\times$). The package delivering less is *expected*: it applies **source-limited** policy effects, not a uniform 30% intake cut.

Limitations

Limitations and interpretation

- **Deaths delayed, not averted.** Case fatality is held at baseline for public procurement, FOPL, salt targets, fiscal *and the primary Na/K→SBP LSS*, so these lower incidence and shift mortality later. Only the optional **SSaSS trial-RR benchmark** lowers stroke case fatality (averting some stroke deaths).
- **LSS is benefit-only — no CKD / hyperkalaemia harm channel** (unlike Huang et al. 2026), so LSS benefits are not offset; read as upper-ish where hyperkalaemia risk matters.
- **LSS inputs.** NCD-RisC eligibility is age-standardised 30–79 (not age-specific); uptake/adherence and whole-population coverage are assumptions; the K→SBP evidence is thinner than Na→SBP and additivity is assumed.
- **Fiscal policy is exploratory.** Pass-through, elasticity, taxable-packaged share and substitution are uncertain, and transferring broad-category elasticities to high-sodium foods is contested; low/base/high bracket but do not resolve this.
- **Point estimates.** Input CIs (Euromonitor, fiscal) are shown but *not* propagated into deaths delayed; no probabilistic sensitivity analysis yet.
- **Provisional source shares & procurement.** Nigeria/Cameroon are working estimates (Nigeria a proxy for Cameroon); public procurement uses a provisional adults-only share.
- **Independent-cause structure.** The four causes share one population but are modelled independently.

Discussion and next steps

Pending decisions — for discussion

Implemented since July 23 (no longer open)

LSS — mechanistic Na/K→SBP is now primary; SSaSS trial RRs are a benchmark only (s4/s5). **Fiscal** — explicit tax→price→demand pathway with low/base/high sensitivity (exploratory). **Euromonitor** — country-specific packaged trends.

Still open — for discussion

1. **Fiscal parameters** — tax rate, pass-through, elasticity, 50% taxable-packaged share and substitution are all RTSL-to-confirm. Is the deliberately simple pathway acceptable (fiscal is lowest priority)?
2. **LSS coverage, uptake & harms** — confirm s2 coverage and s4/s5 household uptake; and whether to add a CKD/hyperkalaemia harm channel (currently benefit-only).
3. **Euromonitor** — keep rice/pasta/noodles? what packaged-share ceiling (80% now; Matti used 50% for India)? composition-only, or let packaged growth raise total sodium?
4. **Uncertainty** — propagate input CIs into deaths delayed, or keep point estimates?
5. **Policy framing** — lives-saved maximization vs. a prevention/counterfactual framing (e.g. Ethiopia).

Next steps and timeline

- **Built since July 23** — LSS switched to the mechanistic **Na/K**→**SBP** primary model (SSaSS a benchmark); fiscal rebuilt as an explicit exploratory pathway with low/base/high sensitivity; Euromonitor moved to country-specific trends; ramp-up 2026\$→\$2030 confirmed.
- **Still pending** — whether to model LSS/CKD harms; whether to propagate input uncertainty.
- **Add the economic layer** — policy costing plus benefit–cost outputs (NPV, BCR) and DALYs; for fiscal, tax revenue and cost-effectiveness (as in Saxena) if it is prioritized.
- **Primary deliverable** — the **BP reinvestment proposal** memo (tables + figures), target **August 2026**.
- **Paper** is secondary; coordinate with Matti on the Bhat/Marklund and Euromonitor inputs to avoid scooping the forthcoming paper.

Discussion

With the Na/K LSS and the explicit fiscal pathway now built, where should we push hardest before the reinvestment deliverable — the pending data, LSS coverage/uptake and possible CKD harms, or the economic layer?

Appendix

Appendix: per-intervention methods

The following slides document each intervention's **evidence base** and the **applied** η used in the model. Applied values are produced by `reports/report.RMD` and `mirror code/07_run_interventions.R`; study effect sizes are cited from source.

Also in this appendix: documented source-share values (Table M1) and the per-scenario η decomposition by country.

A. Public food procurement

Target source: public (institutional meals). **Applied η :** 20% of the public source.

- **Evidence:** Gase et al. (2011) — modeling study, LA County. Weighted **-233 mg/day** ($\approx 7\% \approx \$20.6\%$ of public-meal sodium).
- **Source adjustment:** for a country with share x of sodium from institutional meals, population impact $= 0.20 \times x$. Worked: Viet Nam $\$ = 0.20 \times 1.8\% = \$ 0.0036$.
- **Share UNDER REVISION (provisional):** the flat 1.8% public share assumed **school coverage**; this model is **adults-only**, so the share is being revised *downward*. This run used the stale 1.8% (code ran with `ADULT_POPULATION_ONLY = FALSE`, printing a loud caveat); RTSL owes the adults-only per-country shares. Public-procurement and full-package results are therefore **provisional**.
- **Caveats:** modeled (not observed); US-specific setting; small affected population share.

B. Behavior change / mass media

Target source: discretionary + packaged. **Applied η :** 1% to each.

- **Evidence:** Do et al. (2016) — COMBI Vietnam, 5% fall in mean sodium after a 1-year multi-component campaign. Collins et al. (2014) — UK economic evaluation, 2% [1–5%].
- The estimate was revised toward a UK basis; a conservative 1% is applied to each of discretionary and packaged.

Open question

RTSL funds **policy change, not mass media for behavior change** (Laura). Keep this lever in the model, or drop it?

C. Low-sodium salt substitutes (LSS)

Primary: mechanistic **Na/K** → **SBP** → **CVD** (75% NaCl / 25% KCl), benefit-only. **Benchmark:** SSaSS direct stroke RRs (s4/s5 only).

- **Na channel:** discretionary sodium cut by KCl fraction $\times$ adherence 0.92 $\times$ uptake 1.00 (= 23%); Filippini Na→SBP (2.8 raised-BP / 1.0 normal-BP mmHg/g); WHO 2 g/day floor.
- **K channel:** K added stoichiometrically → excretion (39.1 mg/mmol; \$÷\$1.3) → Filippini U-curve (Filippini, Naska, et al. 2020); baseline-Na modulated; \$×\$0.50 in normotensives. Baseline K: Reddin 2023 (Reddin et al. 2023). Mean Δ SBP (s2) 1.27 mmHg (= 0.33 Na + 0.94 K).
- **Scenarios:** s2 whole population (50%); s4 diagnosed HTN (15%); s5 treated HTN (10%) — s4/s5 from NCD-RisC 2019; aligned with Huang et al. scenarios 2/4/5 (Huang et al. 2026).
- **SSaSS benchmark (s4/s5 only):** stroke incidence RR 0.90 and a conditional case-fatality multiplier 0.856 (Neal et al. 2021); a separate pathway, never combined with Na/K.

LSS model: parameters, reference values, and sources

$$\Delta SBP^{LSS} = \Delta SBP^{Na} + \lambda_K \Delta SBP^K, \quad \Delta Na = Na_0 s_D q_{KCl} a u c$$

LSS composition, exposure, and sodium pathway				Potassium pathway and combination			
Term	Definition	Value	Source	Term	Definition	Value	Source
Na_0	Baseline sodium intake	Country-specific, g/d	Model input	K_0	Baseline potassium intake	Country-sex specific	Reddin et al. 2023
s_D	Share of sodium from discretionary salt	Country/year-specific	Sodium-source data	M_K	Potassium mass per mmol	39.1 mg/mmol	Molecular mass
q_{NaCl}	NaCl mass fraction of LSS	0.75	SSaSS (Neal et al. 2021)	ρ_K	Intake-to-24h urinary excretion ratio	1.3	Filippini (Filippini, Naska, et al. 2020)
q_{KCl}	KCl mass fraction of LSS	0.25	SSaSS (Neal et al. 2021)	r_{Na}	Na mass per g NaCl	$22.99/58.44 = 0.393$	Stoichiometry
u	Uptake among reached population	1.00	Assumption	r_K	K mass per g KCl	$39.10/74.55 = 0.524$	Stoichiometry
a	Adherence: fraction of salt use replaced by LSS	0.92	SSaSS yr-5 use (Neal et al. 2021)	$f_K(x)$	Achieved K excretion → SBP curve	U-shaped; ref. 90 mmol/d	Filippini (Filippini, Naska, et al. 2020)
c	Population coverage / eligibility	s2: 0.50; s4/s5: country-sex specific	s2 assumption; NCD-RisC 2019	m_{Na}	Modification of K effect by baseline Na	0.8 / 1.0 / 1.2	Filippini/Huan based assumption
η_{LSS}	Fraction of discretionary Na displaced, $q_{KCl} a u$	$0.25 \times 0.92 \times 1 = 0.23$	Derived	γ_N	K-effect multiplier in normotensive BP bins	0.50	Assumption
β_{Na}	SBP reduction per g/d Na reduction	2.8 raised BP; 1.0 normal BP	Filippini (Filippini, Malavolti, et al. 2021)	λ_K	Weight on K-mediated SBP effect	1.00	Additivity assumption
Na_{min}	Minimum sodium intake allowed	2 g/d	WHO/model constraint	$RR_{10,c}$	CVD RR per 10-mmHg SBP reduction, cause c	Cause-specific	Ettehad (Ettehad et al. 2016)

D. Front-of-pack labeling

Target source: packaged (estimate already from packaged-food purchases). **Applied η :** 13.8% of packaged.

- **Evidence:** Taillie et al. (2024) — Chile's mandatory warning-label law (observed). **−73.3 mg/capita/day (−13.8%)** in sodium purchases (2,844 households, 2013–2019).
- **Caveat:** purchases-to-intake translation is uncertain.
- **Alternative:** Mexico found $\approx -7\%$ in packaged sales (≈ -38 mg/capita/day).

E. Mandatory salt targets (reformulation)

Target source: packaged. **Applied η :** **20% of packaged** (conservative; 15% floor for sensitivity).

- **Policy:** South African mandatory maximum sodium limits (2013; targets implemented 2016 and 2019).
- **Evidence:** Strauss-Kruger et al. (2022) — African-PREDICT, **−15.2%** sodium; baseline/follow-up straddle the target dates, so likely an **underestimate**.
- **Alternatives:** (Gaziano et al. 2025) −8.4% (rural, spot urine); (Charlton et al. 2021) −18.4%; UK (He, Pombo-Rodrigues, and MacGregor 2013; Song et al. 2023); Mexico modeling (Muciño-Sandoval, Barrientos-Gutiérrez, et al. 2025).

F. Fiscal policy / taxation (exploratory)

Target source: packaged. **Applied η (base): 5% of packaged — exploratory, lowest priority.**

- **Anchor:** Saxena et al. (2025) — Philippines, modeled 20% tax on unhealthy packaged foods; ≈ 100 mg/day ($\approx 10\%$) sodium at year 20.
- **Pathway (base):** $20\% \text{ tax} \times 100\% \text{ pass-through} = 20\% \text{ price rise} \times \text{elasticity } -0.50 \Rightarrow 10\% \text{ fewer taxed products} \times 50\% \text{ taxable share} \Rightarrow 5.0\% \text{ of packaged sodium.}$
- **Sensitivity** (packaged-sodium reduction): 0.7% (low) / 5.0% (base) / 12.6% (high).
- **Alternatives / evidence:** India (Roche et al. 2026); Australia (Davies et al. 2026; Cobiac et al. 2017); broad-category price elasticities (Green et al. 2013); systematic review (Dodd et al. 2020) (1–11% reduction).

Appendix: sodium source shares — documented values

Country	Discretionary	Packaged	Restaurant	Public	Inherent
Viet Nam	0.247	0.148	0.497	0.018	0.089
Philippines	0.331	0.445	0.104	0.020	0.100
Bangladesh	0.769	0.065	0.046	0.020	0.100
China	0.622	0.181	0.077	0.020	0.100
Ethiopia	0.818	0.018	0.044	0.020	0.100
India	0.769	0.065	0.046	0.020	0.100
Malaysia	0.331	0.445	0.104	0.020	0.100
Thailand	0.244	0.146	0.490	0.020	0.100
Nigeria	0.231	0.415	0.205	0.050	0.100
Cameroon	0.231	0.415	0.205	0.050	0.100
default	0.457	0.233	0.186	0.026	0.099

Fractions of total intake, renormalised to 1.0. The `default` row (mean of the ten) applies to any country without its own row.

These are the **2024** shares; the packaged share then grows on the Euromonitor trend (next slide).

Appendix: packaged-food trend (Euromonitor)

The packaged *share* follows a **country-specific** Euromonitor 2025–2030 prediction — a pre-combined per-country growth rate (with 95% CI) over the sodium-relevant categories — grown to a 2030 end-state and **held flat after**. Total intake is unchanged; only *where* sodium comes from shifts. Median growth $\approx 2.9\%/yr$ (range 0.1%–4.3%), capped at 80%.

Country	Growth g (%/yr)	95% CI	Packaged base	Packaged 2030	Packaged 2050
Philippines	3.93	3.82–4.03	44.5%	54.0%	54.0%
Nigeria	4.29	4.17–4.41	41.5%	51.2%	51.2%
Cameroon	2.68	2.34–3.03	41.5%	47.3%	47.3%
Malaysia	0.96	0.86–1.06	44.5%	46.7%	46.7%
default	NA	—	23.3%	23.3%	23.3%
China	1.62	1.46–1.77	18.1%	19.6%	19.6%
Thailand	3.95	3.83–4.06	14.6%	17.7%	17.7%
Viet Nam	2.01	1.97–2.05	14.8%	16.4%	16.4%
India	4.03	3.99–4.07	6.5%	7.9%	7.9%
Bangladesh	3.05	2.66–3.45	6.5%	7.6%	7.6%
Ethiopia	0.06	-0.27–0.39	1.8%	1.8%	1.8%

Source: Euromonitor (Euromonitor_trends_RTSL_v002.xlsx, Marklund). Category selection (incl. whether to keep rice/pasta/noodles) and the packaged-share ceiling are RTSL-to-confirm.

Appendix: η decomposition by scenario and country



Per-scenario `salteff` for every country, stacked by source contribution (note free y-axes). Fiscal now contributes a 5% reduction of the packaged source (provisional).

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